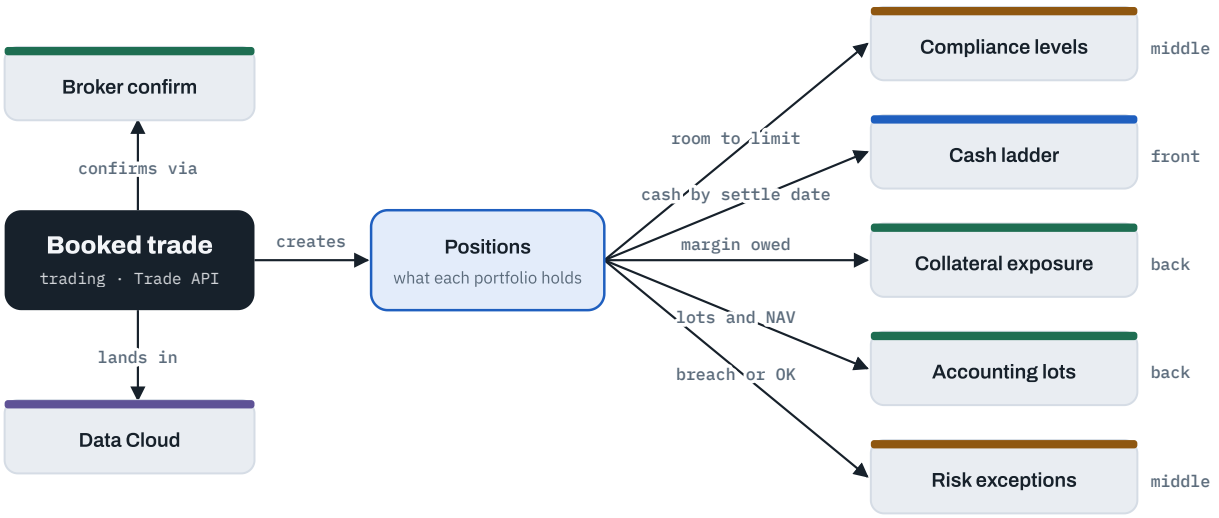
LESSON 1 OF 11

What Aladdin is

Aladdin is the software BlackRock built to run its own investment business, then sold to other institutions. One system holds what each portfolio owns, every order and trade, the rules each portfolio must obey, the cash it has available, its accounting records and its risk numbers. **The product's value is that one booked trade updates every one of those views at once.**

The investment process in six roles

A client hands money and a **mandate** to an asset manager. A **portfolio manager** decides what to buy or sell and raises an **order**. A **trader** executes it with a **broker**. A day or two later the trade **settles**: cash and securities change hands at the **custodian**, the bank that holds the client's assets. **Accountants** record each purchase as a **lot** and compute the fund's daily price, the **NAV**. **Compliance and risk teams** check every step against the mandate and regulation. Aladdin is the system the investor-side roles share; custodians and fund administrators connect to it through the Aladdin Provider network rather than working inside it.



One trade, six consequences. The holdings it changes are read by the compliance check (how close each rule is to its limit), the cash forecast by settlement day, the collateral owed on derivatives, the accounting lots and NAV, and the risk rules. The trade is also confirmed with the broker and copied to the client's Data Cloud. Each arrow is a dependency visible in the API specs; the coloured bands are the office tier that owns each view.

54 / 301
public APIs and operations this guide is built from

11
API domains, one per business function

~100
front-end apps, with thousands of services behind them

7,000
people in the Aladdin organisation, over 4,000 of them engineers

$1BN+
technology business, revenue up 13% in Q2 2026

Who uses it

Asset managers, asset servicers and fund administrators, banks and brokers, corporate treasuries, insurers, pension funds, private-markets managers and investors, and wealth managers. Sold only to institutions, in BlackRock's words "institutional, wholesale, qualified and professional" clients; the retail investors whose portfolios wealth managers run are on the platform without being its customers. BlackRock quotes hundreds of clients across 70 countries.

How it is delivered

Hosted software. Each client gets its own Aladdin environment (a tenant), but every environment runs the same code and sits under one control plane, so BlackRock pushes the same software to all clients. BlackRock staff can be granted access into a client environment as "provider" users. Hosted on Microsoft Azure today, with an AWS option expected in late 2026.

How this guide reads it

From the 54 machine-readable API descriptions (OpenAPI specs) in BlackRock's public SDK packages, cross-checked against about 120 transcripts of BlackRock talks, podcasts, webinars and client videos. Spec-derived names are flagged as such where BlackRock speakers never use them.

How to read this. Lessons 2 to 3 explain what the platform does. Lessons 4 to 6 explain how it is built and are written for engineers and technical product people; a non-engineer can skim them. Move between lessons with the tabs, the arrow keys, or the buttons at the foot of each lesson. Click the nodes in the diagrams.

Lifecycle →

LESSON 2 OF 11

The investment lifecycle Aladdin covers

Each stage hands a record to the next, and the hand-off is a state change on that record. Click a stage to see the record it owns, the states it moves through, and the APIs that move it.



The coloured band on each stage is its office tier: blue front office (decides and acts), amber middle office (controls), green back office (settles and accounts). The word under each arrow is what one stage hands to the next.

← Previous stage

Next stage →

Front office `investment research`

1. Research

Analysts record their thinking: templated notes, logged meetings with companies, who covers which issuer, watchlists, and stored screening rules that alert when a security meets them.

> **Why it is built this way.** Research is where an idea starts, and the platform keeps it attached to the issuer or security it is about, so a later order can be traced back to the note that justified it.

Screening criteria are real expressions evaluated over security identifiers, such as ((price > 55 OR oas = 75) AND market_cap < 1mm). Notes carry permission groups that expire, so research can be shared for a bounded time.

Record ResearchNote, Engagement, Watchlist, Criterion

States `draft` → `published`

PUBLIC APIS

`ResearchNote v2` `Engagement` `AnalystCoverage` `Watchlist` `Criterion`

Open investment research in the explorer →

← Overview

Domain explorer →

LESSON 3 OF 11

Domain explorer

The API layer is called Aladdin Graph, and every API in it has an address of the form `agraph.<domain>.<segment>.<group>.<version>.<API>`. The domain is the business function that owns it. Pick a tier or a domain, filter by stability, or search all 54 public APIs.

All tiers | Front office | Middle office | Back office | Platform

SEARCH order, cash, broker, risk...

STABILITY any

accounting 4 APIs · 8 ops | analytics 9 APIs · 53 ops | clients 1 APIs · 7 ops | compliance

All domains

54 of 54 APIs

Click a domain on the left for what it covers, or search across everything.

PortfolioConfiguration API v1 beta 3 ops LRO accounting

`/api/accounting/configuration/attribute/portfolio/records/v1/`

Configurations API for Aladdin Accounting allows you to access accounting configuration attributes for process types that portfolios are setup on. Data can be sourced in aggregate and...

CompositeMembership API v1 alpha 1 op accounting

`/api/accounting/configuration/composite-membership/v1/`

Membership of portfolios in performance composites, with entry and exit dates and reasons.

AborLot API v1 alpha 1 op accounting

`/api/accounting/transactions/abor-lot/v1/`

Accounting-book-of-record lots, retrieved from the ledger summary table, with the parallel OBOR lot id and open and close transaction ids.

Valuations API v1 beta 3 ops accounting

`/api/accounting/workflow/valuations/v1/`

In an accounting workflow, valuations are generated for various processes at the portfolio/portfolio group level. This service can be used to retrieve and filter for accounting...

Climate API v1 stable 2 ops analytics

/api/analytics/data/climate/v1/

The Aladdin Climate Data API exposes physical, transition, and temperature alignment analytics as of a given date. Users can retrieve data for selected entity types by specifying the...

EsgData API v1 stable 3 ops analytics

/api/analytics/data/esg/v1/

The ESG Data API offers a centralized source of ESG data and meta data across multiple vendors. The API retrieves ESG data by asset and issuer from multiple vendors, providing the data...

EvaluatorAnalytics API v1 stable 4 ops LRO analytics

/api/analytics/oversight/governance/v1/

Trigger Rule evaluations for Rules created and subscribed to within Risk Radar.

RiskConfig API v1 stable 5 ops analytics

/api/analytics/oversight/governance/v1/

Retrieve, update, and create configurations which drive Risk Governance behaviours and Risk Radar UI choices.

RiskCustomEvaluationMetric API v1 stable 4 ops LRO analytics

/api/analytics/oversight/governance/v1/

Upload or retrieve custom evaluation metrics to be used for rule evaluation in Risk Radar.

RiskException API v1 stable 6 ops LRO analytics

/api/analytics/oversight/governance/v1/

Retrieve, update, or create Exceptions as surfaced in Risk Radar.

RiskRule API v1 stable 20 ops LRO analytics

/api/analytics/oversight/governance/v1/

Retrieve, update, and create Rules and Rule Subscriptions as surfaced in Risk Radar.

RiskTask API

v1 · sunset · 1 op · analytics

`/api/analytics/oversight/governance/v1/`

Retrieve Tasks, as surfaced in Risk Radar, which are aggregates that comprise of related Exceptions, Rules, and Workflow items.

RiskWorkflow API

v1 · stable · 8 ops · LRO · analytics

`/api/analytics/oversight/governance/v1/`

Retrieve, update, and transition Workflow items as surfaced in Risk Radar.

EnrichedCapitalFlow API

v1 · stable · 7 ops · clients

`/api/clients/capital-flows/enriched-capital-flow/v1/`

Capital flows are the cash and asset subscriptions coming into a fund and the cash and asset redemptions going out of a fund (e.g., client contributions, withdrawals, and initial funding...

ComplianceRule API

v1 · alpha · 1 op · compliance

`/api/compliance/state/compliance-rule/v1/`

Compliance Rules are used to automatically monitor whether a fund adheres to a regulation, client mandate, or internal guideline.

ComplianceRuleAssignment API

v1 · alpha · 4 ops · compliance

`/api/compliance/state/compliance-rule-assignment/v1/`

Rule assignment assigns compliance rules for portfolio.

Level API

v1 · stable · 1 op · compliance

`/api/compliance/state/level/v1/`

Retrieves start-of-day and intraday Compliance levels across portfolio and portfolio groups.

Violation API

v1 · alpha · 3 ops · compliance

`/api/compliance/state/violation/v1/`

Retrieve and Create Compliance Violations

SecurityCreation API

v1 · stable · 15 ops · data

`/api/data/reference-data/asset/asset-creation/v1/`

This service is used to create CDS, CDX, Equity Equity, Equity Option, Futures, FX NDF, FX SPOT, FX FWRD, FX CSWAP, FX Option, Swap, Swaption, CASH/TD, CASH/REPO, ARM/TBA and MBS/TBA...

CorporateAction API

v1 · beta · 2 ops · investment operations

`/api/investment-operations/asset-lifecycle/corporate-action/v1/`

A corporate action is an event triggered by a public company that changes an equity or fixed income security issued by the company. There are two main categories - Mandatory and Voluntary.

CorporateActionEntitlement API

v1 · beta · 3 ops · investment operations

`/api/investment-operations/asset-lifecycle/corporate-action/v1/`

Per-portfolio entitlements on a corporate action, with elections that are instructed to the custodian and checked against compliance.

CouponReset API

v1 · stable · 7 ops · investment operations

`/api/investment-operations/asset-lifecycle/coupon-reset/v1/`

Coupon in Aladdin are generated internally by Aladdin's Bulk Rate Mechanism (BRM) module or sourced from external vendors. This API allows for retrieval of coupon reset records based on...

PrincipalInterestFactor API

v1 · stable · 9 ops · investment operations

`/api/investment-operations/asset-lifecycle/principal-interest-factor/v1/`

Principal and Interest Factors (PIFs) generally represent the amount of payment per 1000 of Original Face currency units value you currently hold in the given asset, and are used in...

MiscellaneousCashflow API

v1 · beta · 8 ops · investment operations

`/api/investment-operations/cash-flows/miscellaneous-cashflow/v1/`

Create, update and cancel miscellaneous cash, with or without a standard settlement instruction, and send money out through the transmission framework once confirmed and reviewed.

CollateralStatement API

v1 · beta · 3 ops · LRO · investment operations

`/api/investment-operations/collateral-management/collateral-statement/v1/`

Collateral statements in Aladdin are generated on a daily basis by Aladdin's margin calculators referencing legal agreement set up in DECO, security details, pricing and position data....

AccountInfo API

v1 · alpha · 2 ops · investment operations

`/api/investment-operations/reference-data/account-info/v1/`

Account types of an organisation, filterable and searchable by name.

Audit API v1 alpha 1 op investment operations

`/api/investment-operations/reference-data/broker/v1/`

Operations to retrieve audit data for broker entities.

Broker API v1 alpha 7 ops investment operations

`/api/investment-operations/reference-data/broker/v1/`

The other party to a trade. Brokers carry a ticker, type, issuer id, legal entity and business purpose; batch create, update and validate up to 100 at a time.

BrokerConfirmRouting API v1 alpha 5 ops investment operations

`/api/investment-operations/reference-data/broker/v1/`

Instructions for delivering trade information to a broker desk after a trade is booked, so it can be confirmed.

BrokerDesk API v1 alpha 6 ops investment operations

`/api/investment-operations/reference-data/broker/v1/`

Each broker has at least one desk; the desk on a trade designates the group and contact the transaction was done with.

BrokerExternalAlias API v1 alpha 6 ops investment operations

`/api/investment-operations/reference-data/broker/v1/`

Maps an Aladdin broker to the aliases by which it is known outside Aladdin.

CounterPartySettlementInstruction API v1 alpha 9 ops investment operations

`/api/investment-operations/reference-data/broker/v1/`

Settlement instructions for a broker or account, with settlement parties, review state and soft delete.

CounterPartySettlementInstructionTemplate API v1 alpha 1 op investment operations

`/api/investment-operations/reference-data/broker/v1/`

Settlement instruction templates by delivery method.

Contact API v1 alpha 5 ops investment operations

/api/investment-operations/reference-data/contact/v1/

Contact defines the Recipient details to which we want to communicate with external parties like Custodians/Brokers

AnalystCoverage API

v1 · alpha · 9 ops · investment research

/api/investment-research/analyst/coverage/v1/

The Analyst Coverage API enables users to assign, transfer, or remove analyst coverage assignments for one or more entities. It also support discovery of coverage for an analyst or...

Engagement API

v1 · alpha · 6 ops · investment research

/api/investment-research/content/engagement/v1/

Records of meetings with companies and issuers: attendees, topics, momentum, KPIs, draft and published states.

ResearchNote API

v2 · alpha · 8 ops · investment research

/api/investment-research/content/research-note/v2/

Templated research notes with custom fields, permission groups with expiry, and tags to issuers, assets and strategies.

Criterion API

v1 · alpha · 9 ops · investment research

/api/investment-research/surveillance/criterion/v1/

Stored screening expressions, for example ((price > 55 OR oas = 75) AND market_cap < 1mm), evaluated against securities with scheduled alerts.

Watchlist API

v1 · alpha · 9 ops · investment research

/api/investment-research/surveillance/watchlist/v1/

Create, modify, delete, and retrieve watchlists; Add and remove securities from watchlist.

Permission API

v1 · stable · 2 ops · platform

/api/platform/access/permission/v1/

Permissions have a functional area, action and sensitivity grade. They attach to user groups, never directly to users. Not intended for runtime authorisation.

UserGroupPermission API

v1 · stable · 3 ops · platform

/api/platform/access/permission/v1/

Add and remove permissions on a user group.

UserGroup API v1 stable 5 ops platform

`/api/platform/access/user-group/v1/`

Hierarchical user groups with node types for groups that hold users and internal groups that hold other groups.

UserGroupMember API v1 stable 3 ops platform

`/api/platform/access/user-group/v1/`

Add and remove users from a user group.

User API v1 alpha 5 ops platform

`/api/platform/identity/user/v1/`

An Aladdin user is a person or system account: client employees, provider (BlackRock) employees granted access, and bot accounts, with alternate ids mapped to the client's own HR identifiers.

AdcDataset API v1 alpha 11 ops LRO platform

`/api/platform/studio/adc/adc-dataset/v1/`

Manages Datasets in Aladdin Data Cloud (ADC). Used by Studio's ADC Admin Center.

PortfolioGroup API v1 stable 11 ops portfolio

`/api/portfolio/configuration/portfolio-group/v1/`

The portfolio group hierarchy, with ancestor, descendant, parent and child queries and permissioned membership changes.

RestrictedAsset API v1 alpha 6 ops portfolio

`/api/portfolio/configuration/restricted-asset/v1/`

Restricted assets serve as a safeguard and are used prevent assets from being traded in Aladdin. The Restricted Assets API can be used to list, create, update, upsert, filter and...

PortfolioToolkit API v1 stable 6 ops portfolio

`/api/portfolio/setup/portfolio-toolkit/v1/`

Retrieve, filter and search portfolios and portfolio slices. The full Portfolio entity carries 850 attributes; slices expose summaries and dynamic attributes.

CashLadder API

v2 · stable · 2 ops · portfolio management

`/api/portfolio-management/cash/cash-ladder/v2/`

Cash Ladder provides a settlement date based ladder of cash balances for all the exposure currencies for a given portfolio group and cash date.

CIDReader API

v1 · stable · 7 ops · portfolio management

`/api/portfolio-management/setup/cids/v1/`

Read custom investment dataset definitions, schemas, hierarchies and entities. CIDS does not transform the data; the writer owns it.

CIDWriter API

v1 · stable · 10 ops · portfolio management

`/api/portfolio-management/setup/cids/v1/`

Create and activate dataset definitions and schemas, then load entities under approved leaf datasets, up to a configured maximum.

Strategy API

v1 · alpha · 9 ops · portfolio management

`/api/portfolio-management/target/strategy/v1/`

Named investment strategies referred to by orders and trades, with soft delete by expiry date and wildcard search.

Order API

v1 · stable · 7 ops · LRO · trading

`/api/trading/order-management/order/v1/`

Filter, post or cancel orders. An order is a directive from a portfolio manager to the trading desk to execute a particular investment decision.

Trade API

v2 · alpha · 7 ops · LRO · trading

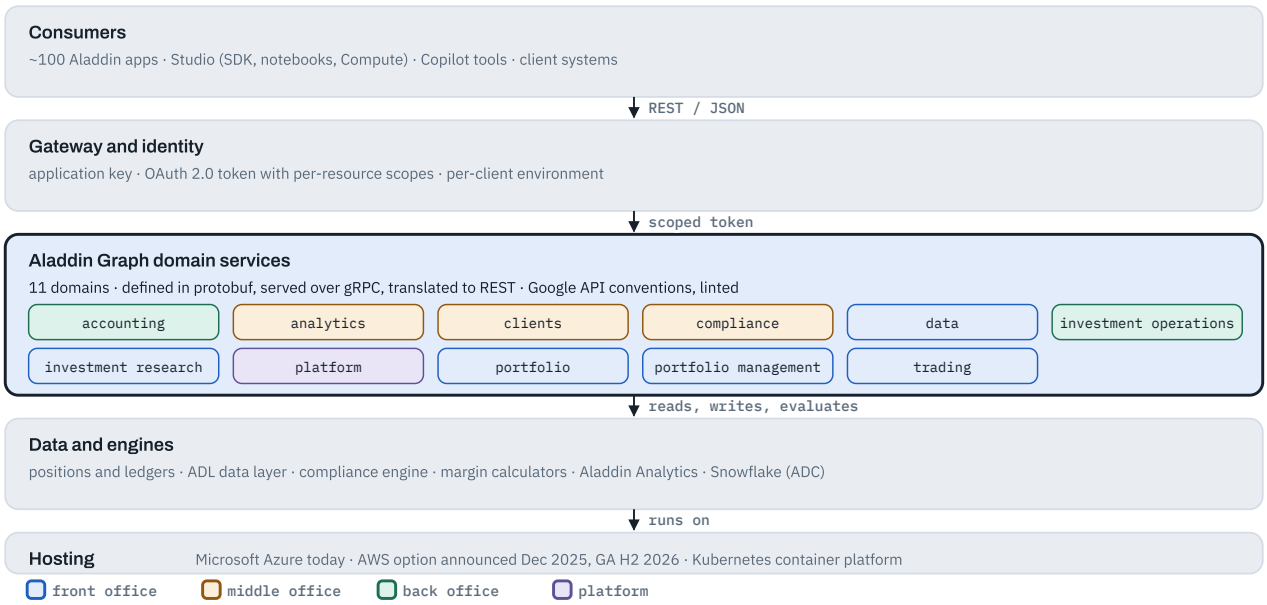
`/api/trading/trade-processing/trade/v2/`

Post, cancel, filter and retrieve block trades with their allocations, broker quotes, settlement details and trade relationships.

LESSON 4 OF 11

How it is layered

Every caller, whether a BlackRock app, a client script or Copilot, enters through the same gateway and the same domain services. There is no privileged path. The domain split is deliberate: Aladdin began as literally one database, trading was later carved out onto its own store, and BlackRock now says the domain, not the database, is the source of truth for its data. Click a layer to see what the specs say about it.



The specs describe the middle three layers directly. The top and bottom come from BlackRock's announcements and job postings.

The service layer the specs describe

Protobuf is Google's schema language for describing messages and services; gRPC is the binary protocol that serves them. Every Aladdin spec carries the fingerprints of both (gRPC status types, update masks, page tokens, LongrunningOperation, enums named `FOO_UNSPECIFIED`), which means the services are defined once in protobuf and translated to ordinary REST and JSON at the edge. URLs follow `/api/<domain>/<segment>/<group>/v1/`. Inline comments such as `api-linter: aladdin::9016 … aip.dev/not-precedent` show BlackRock runs Google's API linter plus its own rule set, and that exceptions need a written reason. The commonest reason given is that changing a URI "will break all citizen developers code", BlackRock's term for finance staff who write some code.

← Domain explorer

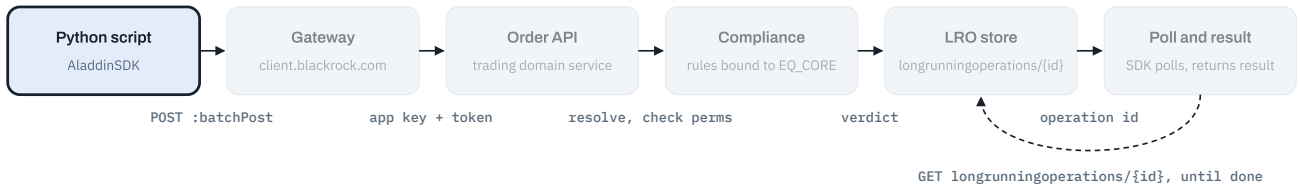
One request →

LESSON 5 OF 11

Anatomy of one request

Follow a portfolio manager's order from a Python script to a booked order with a compliance verdict. Watch for three things: two separate permission checks, compliance running inside the order post rather than as its own call, and the HTTP response coming back before the work is done.

1. Python script | 2. Gateway | 3. Order API | 4. Compliance | 5. LRO store | 6. Poll and result



A portfolio manager's script posts a batch of orders through the SDK.

1. The call

synchronous

The SDK validates the request body, attaches the application key and an OAuth bearer token, and sends one HTTP request. Configuration comes from layered settings files; run mode is local, compute or test.

```
POST /api/trading/order-management/order/v1/orders:batchPost
VND.com.blackrock.API-Key: <app key from Studio>
Authorization: Bearer <token with scope …Order:write>

{"postOrderConfig":
{"whatIf":false,"runCompliance":true,"complianceTimeout":"30s"},
 "orders":[{"assetReference":{"isin":"US0378331005"},"transactionType":"BUY",
   "orderDetails":[{"portfolioReference":
{"portfolioTicker":"EQ_CORE"},"quantity":10000,"strategyName":"Q4 rebalance"}]}]}
```

← Back | Next hop →

LESSON 6 OF 11

Design conventions you can see in every spec

The same dozen patterns repeat across all 54 APIs, which is itself the finding: the platform is governed, not grown. Each card gives the pattern, the evidence in the documents, and why it matters to someone building on or running the platform.

Resources plus custom methods −

PATTERN

Google-style URIs: a collection, then a colon verb for anything that is not plain create, read, update or delete.

EVIDENCE IN THE SPECS

`POST /orders:filter, /orders:batchPost, /order:cancel, /brokerDesks:undelete, /workflows:listRevisions. 36 filter, 17 batchCreate and 13 batchUpdate methods across the public set.`

WHY IT MATTERS

A client learns one grammar once and can guess the shape of an API they have never seen. Filtering by POST with a criteria body allows large, structured queries that would not fit in a URL.

Long-running operations −

PATTERN

Bulk and heavy calls return an operation id; the caller polls for done, error and response.

EVIDENCE IN THE SPECS

`GET /longrunningoperations/{id} in 10 APIs, 35 operations flagged x-aladdin-lro. Order and trade batch posts, risk evaluation, ADC dataset registration.`

WHY IT MATTERS

A request that would take minutes does not hold a connection open or time out at the gateway. The cost is that callers must handle "not yet", and retries have to be safe to repeat.

Stability labels and independent versions +

Linter-enforced design with written exceptions +

Protobuf under REST +

Governance in the data model +

Bitemporal records and audit +

Two-layer authorisation +

Bulk by default, with dry runs +

Multi-tenant by construction +

Client extensibility in the schema +

Internal expression languages +

← One request

Products →

LESSON 7 OF 11

The product family

The core products are entry points onto one set of positions and portfolios, so a client that starts with Aladdin Risk and later adds the full platform does not reload its data. eFront is the exception: a separate private-markets suite with its own book of record that is integrated with Aladdin. BlackRock's strategic frame for all of it is the "whole portfolio", public and private assets in one view. The APIs in this guide belong to the core; Studio is how clients build on it.

Aladdin (core platform)

Portfolio management, risk analytics, trading, compliance, operations and accounting on one investment book of record that now spans public and private assets: BlackRock rebuilt private credit natively in Aladdin with facility-level data and over ten private-markets transaction types, and reuses its SWIFT connectivity for private capital calls. The 11 API domains in this guide belong here. Press coverage of the AWS deal calls the full package "Aladdin Enterprise"; BlackRock speakers just say Aladdin.

Whole Portfolio

BlackRock's name for the public-plus-private strategy rather than a separate product: one book of record, risk model and reporting view across asset classes. BlackRock says half of its private-markets sales are now whole-portfolio solutions and quotes 2,000-plus private-markets risk factors monitored daily. Clients' allocations have moved from 60/40 toward 50/30/20 with privates in the mix.

Aladdin Risk

The risk analytics engine offered on its own: factor exposures (BlackRock quotes a model of about 3,000 risk factors, with 5,000 monitored daily), value at risk, stress testing and Market Driven Scenarios built by the Risk and Quantitative Analysis group. Its outputs are the metric source for the Risk Governance rules in the analytics domain. Explore is the flagship analysis and reporting app clients use on top of it. "Aladdin Studio for Aladdin Risk" packages programmatic access to the same analytics with a governance console for access management and usage monitoring.

Aladdin Studio

The data and developer platform. Its head of engineering describes three pillars: everything Aladdin can do exposed as APIs and events, Aladdin Data Cloud on Snowflake, and developer tools such as sandboxes and research environments. Concretely that is the REST APIs (named Aladdin Graph in the specs) behind a login-only developer portal, the open-source Python SDK and plugins, hosted Compute for Python, and Excel and data-exchange tools. Analytics APIs for portfolio and security analysis sit under the same umbrella but are not in the public packages. Studio Developer is its developer-facing half, which BlackRock's product postings call the platform's extensibility layer.

Aladdin Data Cloud

A data-as-a-service "data network" on Snowflake, started around 2019 and launched in 2021 to replace flat files and FTP. BlackRock lands Aladdin data in Snowflake through Fivetran pipelines and shares it into each client's own Snowflake account and cloud, with about 15 minutes of latency; clients bring their own and third-party data in and distribute Aladdin data out to their own ecosystems. BlackRock operates hundreds of Snowflake accounts and tens of thousands of pipelines. The AdcDataset API shows draft, review and released dataset states and ETL configuration.

Aladdin Copilot

Generative AI assistant built with Microsoft on Azure OpenAI and announced in 2023. eFront Copilot reached select clients first in 2024, Aladdin Copilot followed, and BlackRock now says it is available to all Aladdin clients and embedded across the hundred front-end apps. It will not give investment advice or answer outside the platform. See the AI lesson.

Aladdin Wealth

Portfolio analytics, risk and proposals for wealth managers and private banks, built on the same risk model as the institutional platform. Its "Auto Commentary" generative AI tool, announced 2 October 2025 with Morgan Stanley as first implementer, drafts client-meeting commentary from holdings, risk analytics and the firm's own CIO views. An iCapital integration takes advisers from proposal into private-fund subscription.

Aladdin Climate

Physical, transition and temperature-alignment risk analytics under NGFS and IPCC scenarios, covering about 3 million securities and 9,000 ultimate issuers, built by a team of around 100. The Climate API returns metrics per security, deal or issuer under named scenarios and timeframes.

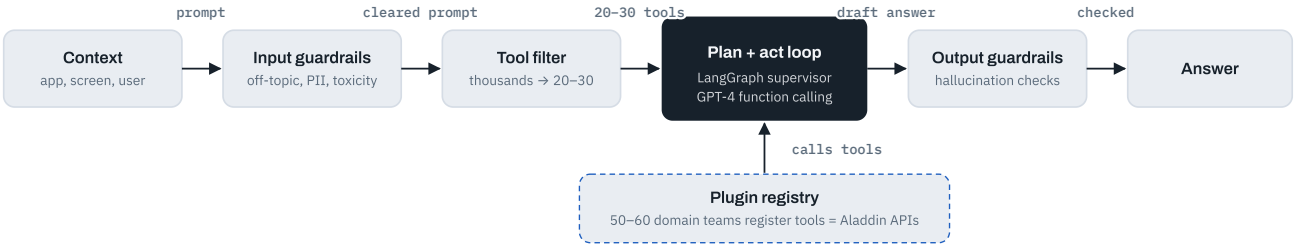
eFront and Preqin

Private markets. eFront, BlackRock's first major technology acquisition (2019), is a modular suite for GPs, LPs and fund administrators covering deal flow, fundraising, fund accounting, investor portals (Investment Cafe), portfolio-company KPIs (Portfolio Monitoring) and the long-standing GP-to-LP data exchange eFront Insight, with its own investment book of record. Preqin (announced 2024, closed March 2025) supplies pre-investment data from about 500 researchers. In February 2026 Preqin data was folded into eFront Insight, and Preqin indices already appear in Explore and Data Cloud.

LESSON 8 OF 11

Where AI plugs in

Aladdin Copilot is an assistant embedded across the apps. When it needs data or wants to act, it calls Aladdin APIs as "tools", the same governed APIs Studio exposes to clients. That is why API quality and AI readiness are one problem for the platform team.



From the LangChain Interrupt 2025 talk by Brennan Rosales and Pedro Vicente Valdez of BlackRock's AI engineering team. The registry holds a couple of thousand tools and agents; access is filtered by environment, user group and application before planning, because more than 40 to 50 tools degrades the planner. BlackRock's own vocabulary is "plug-ins", with "tools" for registered APIs and "custom agents" for the rest.

Federated ownership

The central AI team owns orchestration, guardrails and evaluation. Domain teams onboard by registering existing production APIs as tools or by building custom agents, and every plug-in must come with ground-truth examples. That mirrors the domain split in the API namespace. BlackRock also expects clients to want its orchestrator or plug-ins inside their own copilots, and Aladdin plug-ins inside Microsoft 365 Copilot labelled "powered by Aladdin".

Evaluation-driven development

Synthetic data, subject-matter-expert cases and model-graded pipelines run on every pull request and nightly, with reports per plugin owner. The team compares it to test-driven development. This is recent: in late 2024 BlackRock's head of AI product said there was no universal way to score responses and that getting past proofs of concept was the hard part.

Why it matters for Studio

An agent can only use an API that describes itself well: clear metadata, consistent methods, fine-grained scopes. Studio's head of engineering frames the AI strategy as three-fold: AI in every product, AI as a platform for builders, and AI to help the builder write code. The risk product team says it is refactoring data models so they are "structured for agentic use cases".

← Products

Glossary →

LESSON 9 OF 11

Names that leak through the docs

Internal systems, modules and jargon the API descriptions mention by name. They appear in the API text itself, so an interviewer will recognise them.

Filter terms...

Aladdin Graph (AGraph)
The unified API layer as named in the specs (ids start agraph....). BlackRock speakers just say "the APIs" or "the Studio umbrella".

Studio
The data and developer platform: APIs, portal, SDK, Data Cloud, Compute. Studio Developer is its developer-facing half.

ADC
Aladdin Data Cloud: per-client Snowflake databases of Aladdin data, managed through an Admin Center.

ADL
A data layer the specs reference as the store for ADC dataset repositories and for portfolio search.

Risk Radar
The oversight application where Risk Governance rules, exceptions and workflows are surfaced. The analytics/oversight APIs serve it.

Risk Governance
The risk team's limit framework (rules, subscriptions, exceptions, workflows), distinct from compliance.

DECO
The tool where collateral legal agreements (ISDA, CSA) are set up. Margin calculators read it to generate collateral statements.

Abacus
A compliance engine mode on rules: parallel or live. Suggests shadow evaluation of a new engine.

newcash
The capital flows module. Its permissions (VIEW_NEWCASH, CASH_PAY, CASH_CONF...) gate the EnrichedCapitalFlow API.

CIDS
Custom Investment Datasets: client-owned data loaded into Aladdin in a Category.Class.Group.Dataset hierarchy for use in portfolio management tools.

BRM
Bulk Rate Mechanism, the internal module that generates floating-rate coupon resets.

BQL

An expression language used in filters, order flags and routing rules (bqlExpression, bqlFilter). JQL appears on trade routing.

ABOR / IBOR

Accounting book of record and investment book of record. BlackRock speakers say IBOR; the AborLot spec field is literally named oborLotId, so the two are the same thing.

Explore

The flagship web app for portfolio analysis and reporting: saved workspaces, widgets, drill-down from group to portfolio to asset, stress and what-if panels, performance attribution. Not in the public API packages but demoed in most BlackRock webinars.

Whole Portfolio

BlackRock's frame for one book of record, risk model and view across public and private assets. Canadian pensions call the same idea the Total Portfolio Approach.

Language of the portfolio

Studio engineering's name for the unified symbology and data model that lets Aladdin join market, transaction and third-party data.

Analytics factory / intelligence factory

How Studio's head of engineering describes Aladdin's evolution: from producing the daily risk report (once printed on green paper, the "green package") to driving decisions with data and AI.

Market Driven Scenarios (MDS)

Stress scenarios built by the Risk and Quantitative Analysis (RQA) group from a narrative, calibrated policy-variable shocks and implied factor shocks, shipped to clients who can customise them.

Aladdin Provider

The network of asset servicers and custodians (nine of the largest, JPMorgan since 2016) that exchange gold-copy messages and reconciliation breaks with Aladdin clients.

PIF

Principal and Interest Factor: payment per 1,000 of original face on mortgage and asset-backed securities, used with positions to generate cash flows.

SSI

Standard Settlement Instruction. Cashflows can default an SSI from the broker and then be transmitted.

Transmission framework

The mechanism that sends money out once a cashflow has a confirmer and reviewer.

LRO

Long-running operation. Returned by batch and heavy calls; polled at

Four eyes

Maker-checker approval: one person proposes, a second approves. Named on ADC dataset

/longrunningoperations/{id}.

review and compliance rule assignments.

Gold copy

In the specs, a reference client whose ADC datasets others are reconciled against. In operations, "gold copy messaging" is the authoritative message set exchanged with custodians.

Spec-only names

Aladdin Graph, Studio Developer, Compute, Risk Radar, DECO, CIDS, ADL and BQL appear in the API specs and job postings but are never spoken in any public BlackRock talk; use them knowingly.

Portfolio group

A hierarchy of portfolios and sub-groups. Most APIs accept a group ticker as scope.

invnum

Investment number, part of the trade key with portfolio and trade number.

touchCount

A per-record modification counter used for optimistic concurrency alongside version.

Provider user

In the specs, a BlackRock employee granted access into a client's Aladdin environment. Not to be confused with BlackRock's "user-provider model", meaning it is the first user and the vendor of the same software, or with the Aladdin Provider asset-servicer network.

Citizen developer

Industry term BlackRock adopted (its programme is "citizen development"; some say "embedded engineers") for finance professionals who code. BlackRock counts about 3,000 internally and says that number has probably peaked as natural-language access takes over.

TBA

To-be-announced: a forward trade in agency mortgage bonds where the exact pools are assigned later.

ISDA / CSA

The master agreement for over-the-counter derivatives and its credit support annex, which sets collateral terms.

Access.Internal

A tag on every public spec. Appears to be a visibility classification; its exact meaning is not documented publicly.

Alpha (accounting)

Fields prefixed alpha (alphaLedgerTable, alphaStartDate, alphaPriceGroup) point to the accounting engine behind Aladdin Accounting.

← AI

Quiz →

LESSON 10 OF 11

Test yourself

Click a card to reveal the answer. Then use the questions at the bottom in the interview; each one comes from something in the specs.

What is an Order, in Aladdin's own words?

"A directive from a portfolio manager to the trading desk to execute a particular investment decision." A PM order becomes a trader order when the desk picks it up.

SHOW ANSWER ▾

How does pre-trade compliance reach a client script?

Pre-trade compliance runs inside the order post: runCompliance defaults to true and violations come back in the response. The Compliance APIs manage rules and read levels; they do not check an order.

SHOW ANSWER ▾

Name the two limit frameworks and who owns them.

Compliance (regulatory and mandate rules, run on orders and trades, owned by compliance) and Risk Governance in Risk Radar (metric thresholds on analytics, owned by risk, with its own exceptions and workflows).

SHOW ANSWER ▾

What does the Cash Ladder include beyond settled cash?

Intraday trades, authorised and open orders, confirmed and estimated capital flows, and collateral pledged and held, laid out by settlement date and currency.

SHOW ANSWER ▾

How is Aladdin Data Cloud structured per client?

Each client gets its own Snowflake database. Datasets are tables, views or history views, moving through draft, review (four eyes) and released, with ETL config and reconciliation against a gold-copy client.

SHOW ANSWER ▾

What tells you the services are protobuf and gRPC underneath?

google.rpc.Status, Any with @type, google.type.DateTime, updateMask, pageToken, LongrunningOperation and FOO_UNSPECIFIED enums in every spec.

SHOW ANSWER ▾

How many public APIs are alpha, and why does that matter?

24 of 54. Nearly half the public surface can still change, so graduation criteria and client

What are the two layers of authorisation?

OAuth scopes per resource at the gateway, then Aladdin permission codes (brkrMaint,

migration are live product problems.

SHOW ANSWER ▾

PortGroupMOD, VIEW_NEWCASH) checked inside the service.

SHOW ANSWER ▾

How does Copilot get its tools?

A federated plugin registry: 50 to 60 domain teams register existing production APIs as tools or build custom agents. The central team filters to 20 to 30 tools per query by environment, user group and app.

SHOW ANSWER ▾

What does "one booked trade updates every view" mean concretely?

Level reads positions for room to limit, CashLadder folds in open orders and collateral, CollateralStatement reads positions for margin, AborLot is created from the trade, and RiskRule evaluates analytics computed on the same positions.

SHOW ANSWER ▾

What is CIDS for?

Letting clients load their own datasets (scores, targets, classifications) into Aladdin so screens, rules and reports can use them without exporting to Excel. Aladdin does not transform the data; the writer owns it.

SHOW ANSWER ▾

Which cloud, and what changed in December 2025?

Azure today: the strategy began around 2019, the first phase was a lift-and-shift, and all clients were on Azure by late 2024. In December 2025 BlackRock and AWS announced an AWS hosting option with general availability expected in the second half of 2026.

SHOW ANSWER ▾

What do BlackRock people mean by the whole portfolio?

One investment book of record, risk model and view across public and private assets. It is the strategic frame behind the eFront and Preqin acquisitions, and BlackRock says half its private-markets sales are now whole-portfolio solutions.

SHOW ANSWER ▾

Questions worth asking them

1. Of the public APIs, nearly half are alpha. How does an API graduate to stable, and how do you handle clients who built on alpha when it changes?
2. Compliance and Risk Governance are two limit frameworks with separate rules, exceptions and workflows. Are they converging?
3. What share of Studio usage is BlackRock's own teams versus clients, and what is the path from internal success to external adoption?
4. Is the Copilot plugin registry fed from the same catalog as the Studio developer portal, or are they separate?
5. Where is the line between exposing something as an API, as a Data Cloud dataset, or as an event stream?
6. The specs still cite legacy permission codes under OAuth scopes. Is the goal to retire those, or to keep two layers?
7. What changes for Studio when a client is hosted on AWS rather than Azure?

← Glossary

Sources →

LESSON 11 OF 11

Sources and caveats

Primary evidence

- The 11 `asdk_plugin_*` packages on PyPI, version 2.0.0b3: 54 Swagger 2.0 specs, 301 operations, read in full.
- blackrock/aladdinsdk and aladdinsdk-plugin-builder on GitHub.
- BlackRock Aladdin Studio and APIs pages.
- Aladdin Data Cloud press release, 2021.

Secondary evidence

- AWS partnership announcement, December 2025.
- Microsoft customer story on the Azure migration.
- Summary of a 2025 BlackRock AI engineering talk (third party).
- BlackRock careers postings for Studio Developer product and engineering roles, 2025 and 2026.
- Transcripts: the LangChain Interrupt 2025 Copilot talk, the Snowflake Data Cloud podcast with Aladdin Studio's head of engineering, 42 Never Done and Client Stories podcast episodes, 12 Aladdin webinar replays, 15 Big Little Questions episodes and eFront product videos.
- Q2 2026 results coverage.

What is not here: positions, risk analytics outputs, performance, pricing, benchmarks, execution connectivity, settlement messaging, eFront and wealth APIs. They sit behind the client login to the Studio developer portal; Studio's general manager confirms analytics APIs exist there. Inferences about gRPC, the linter and internal system names come from field names and comments inside the specs, not from BlackRock architecture documents. This page was fact-checked against about 120 transcripts of BlackRock talks, Never Done podcast episodes, webinar replays and client videos; corrections from that pass are incorporated. One widely shared YouTube video titled "Aladdin Asset Management Software" is a demo of an unrelated facility-maintenance product called Aladdin CMMS, not BlackRock.